# Fundamentals of Deep Learning

Lecture Notes

Muxammadrizo Maribjonov

November 3, 2026

## Contents

**1 Introduction** **2**

**2 Neural Networks and the Multilayer Perceptron** **3**

**3 Backpropagation** **4**

**4 Optimization for Deep Learning** **5**

**5 Regularization** **6**

**6 Convolutional Neural Networks** **7**

**7 Recurrent Neural Networks** **8**

**8 Attention and Transformers** **9**

**9 Generative Models** **10**

# 1 Introduction

# 2 Neural Networks and the Multilayer Perceptron

# 3 Backpropagation

# 4 Optimization for Deep Learning

# 5 Regularization

# 6 Convolutional Neural Networks

# 7 Recurrent Neural Networks

# 8 Attention and Transformers

# 9 Generative Models